

When Does Selection Replace Extraction? A Pre-Registered Test of Agent Memory with a Typed Decision Model

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Abstract

Does conversational memory need LLM-extracted facts, or is it enough to select the right raw turns? Published results disagree: extraction-based systems report gains from distilled facts, while recent studies find raw history with good ranking does as well, and disagree about whether ranking matters. We show the disagreement is largely about the context budget. In a pre-registered study on held-out LoCoMo conversations and LongMemEval, raw turns selected by a single call to Jev, a typed decision model, are non-inferior on LoCoMo to an LLM-extraction memory at a tight budget (-0.5 points, one-sided 95% bound -3.0 ; -1.7 to -2.6 under blind human grading) at $3,061\times$ lower write cost, and the result holds under a second answer model. Selection’s value depends on how hard the budget cuts the candidates: reranking adds 17.4 points on LoCoMo and 9.1 on LongMemEval when three of 30 are kept, and 1.5 and 1.1 at generous budgets, where extraction systems are more accurate. Jev selects as accurately as an LLM reranker (non-inferior, bound -2.0) at a third of the latency, and more accurately than a multi-call graph traversal. Reranking lowers correct abstention. Plans, code and all graded answers are released.

1 Introduction

Does conversational memory need LLM-extracted facts? The literature is split. Extraction-based systems report gains from distilling conversations before retrieval: mem0 (Chhikara et al., 2025) extracts facts from each message, the LongMemEval design study (Wu et al., 2025) finds that expanding index keys with extracted facts helps retrieval, and SeCom (Pan et al., 2025) segments sessions and compresses the segments before retrieval. Recent studies find that raw history, ranked well, does

as well or better: SmartSearch (Derehag et al., 2026) retrieves from raw history with a deterministic pipeline and a learned ranking stage, and Fidelity Before Structure (An, 2026) finds verbatim chunks ahead of LLM-extracted artifacts in a controlled comparison. These two also disagree with each other: SmartSearch identifies ranking as the bottleneck, while Fidelity finds that reranking adds little.

This paper explains the disagreement by the context budget, the number of retrieved items the answer model reads. When the budget keeps a few of many candidates, which items are kept decides the answer: selection matters, and a good selector over raw turns can stand in for extraction. When the budget is generous, similarity order already includes most of the evidence, ranking adds little, and extracted facts, which are more compact, are more accurate. We test this under a pre-registered plan, on conversations never used for development: are raw turns with a single reranking call non-inferior to a strong extraction-based memory at a tight, matched budget, and how does the rerank’s value change as the budget grows?

The selector is Jev, TypeSafe’s typed decision model (TypeSafe AI, 2026), which answers a fixed-option question with a probability in one short request. The extraction system is engram v2, which extracts facts with gpt-4o-mini and types, relates and updates them with Jev; it was the most accurate system on our development conversation and was chosen as the comparator because the test could fail against it.

Our contributions:

1. **The context budget explains the published disagreement.** Over similarity search, reranking raw turns adds 17.4 points on LoCoMo and 9.1 on LongMemEval when three of 30 candidates are kept, and 1.5 and 1.1 at $k=20$, where extraction systems are more accurate (§5.3, §6).

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2. **To our knowledge, the first pre-registered non-inferiority test in conversational-memory evaluation**, on held-out conversations, with a second answer model and blind human grading. It bounds what extraction adds at a tight budget: at most 4.7 points under the worst grading we applied, with point estimates of -1.7 to -2.6 points by human grading (§5.1, §5.4).
3. **To our knowledge, the first answer-level, matched-context evaluation of a typed decision model as the selector**. Jev is non-inferior to a gpt-4o-mini listwise reranker (S4, lower bound -2.0) at about a third of the latency, and more accurate than Jev-Mem’s multi-call Jev graph traversal (S2) (§5.2, §5.7).
4. **Diagnostics**. The selection ceiling decomposes into shortlist misses (11.5% of questions) and rerank drops (9.2%), with temporal evidence dropped most at the rerank; and the LLM judge’s leniency interacts with answer length, so judge–human agreement differs by system (§5.4, §5.6, §6).

What is not new: raw turns plus a reranker is a known pattern (Derehag et al., 2026; Wu et al., 2026), and engram v2 is the system of our earlier preprint (Sharma, 2026c); the contribution is the test, the budget explanation and the typed selector, not a new architecture.

2 Related Work

Raw history against extraction. SmartSearch (Derehag et al., 2026) argues that neither LLM structuring at ingestion nor learned retrieval policies are necessary, and ranks raw history with a CrossEncoder and ColBERT fusion stage. Fidelity Before Structure (An, 2026) swaps only the stored representation inside one pipeline and finds verbatim chunks ahead of LLM-extracted artifacts by 15.9 points on LoCoMo (categories 1–3, 699 questions) and 22.0 on LongMemEval-S (500 questions), with gpt-4o answering and a gpt-4o-mini judge giving binary grades. In an external-system anchor (their Appendix D), the official Mem0 package also trails verbatim chunks: 36.6% against 47.9% with a gpt-4o-mini answerer (categories 1–3), and 54.7% against 69.9% with gpt-4o (1,540 questions, categories 1–4). Nano-Memory (Wu et al., 2026) answers from raw turns with retrieval and generation alone; EMem (Zhou and Han, 2025)

builds a strong baseline from near-verbatim discourse units; Zeng et al. (2024) sweep chunks, triples, facts and summaries and find chunk-based and mixed stores strongest on LoCoMo; the Long-MemEval design study (Wu et al., 2025) finds round-level storage best and fact-augmented index keys helpful; and Letta reports 74.0% on LoCoMo for a gpt-4o-mini agent that stores conversation history in files, with no judge stated (Letta, 2025). Our result agrees with this lineage at tight budgets and is smaller and more cautious than Fidelity’s gap, as expected for an extraction system that keeps source quotes. We extend it with a registered non-inferiority margin, held-out conversations, a budget analysis and per-category recall.

Extraction systems. mem0 (Chhikara et al., 2025) extracts facts per message; we test mem0 OSS 2.1.0, and newer mem0 releases report higher, self-reported numbers (Mem0 Engineering Team, 2026). Graphiti/Zep (Rasmussen et al., 2025), A-MEM (Xu et al., 2025), MemGPT/Letta (Packer et al., 2023), EverMemOS (Hu et al., 2026) and Memora (Xia et al., 2026) structure memory with LLM calls at write time.

Typed decisions in memory. Jev-Mem (Jiang et al., 2026a) was the first memory system built on Jev; it uses typed questions for typing, relations, routing, traversal and stopping over a multi-graph store. The AtMem–Jev article (Taghia, 2026) reports that Jev reranking raises ranking metrics. We measure a Jev reranker at the answer level, against an LLM reranker and against Jev-Mem at matched context.

Reranking in conversational memory. SmartSearch finds ranking to be the bottleneck; Fidelity finds reranking marginal. Training-Free Lexical–Dense Fusion (Lysenstøen, 2026) reports an off-the-shelf cross-encoder lowering Hit@1 on conversational queries, and ConvMemory v2 (Pan, 2026) reports gains from a cross-encoder fine-tuned for conversation. Our budget analysis offers one way these findings fit together (§6).

Evaluation validity. Held-out conversation splits of LoCoMo already exist (Yan et al., 2025; Jiang et al., 2026b); our design adds pre-registration and a non-inferiority margin. Same Ranking, Different Winner (Panthi and Abdelfattah, 2026) shows that retrieval credit depends on the stored form; we score shortlist recall on raw turns only. Fidelity reports judge–human agreement of $\kappa = 0.897$ on 100 questions, similar for short and long answers, with a judge instructed to

be strict (their Appendix J.4); with mem0’s lenient LoCoMo judge, we found that agreement depended on the system’s answer style (§5.4, §6).

3 Systems

All systems use gpt-4o-mini to answer, text-embedding-3-small to embed and jev-1.13.0 for every Jev decision. Figure 1 contrasts the write and read paths of T0R, engram v2 and Jev-Mem.

T0R. The write path embeds each turn and stores it as “[date] speaker: text”, with no extraction and no LLM call. The read path takes a 30-turn cosine shortlist and asks Jev, in one request, whether each turn helps answer the question; turns above 0.5 are kept in order of Jev’s probability, followed by a cosine floor of 10 turns. The answer model sees the first k lines.

L0. The same store, read in cosine order with no Jev call.

T0R-LLM. T0R’s store and shortlist, scored by a gpt-4o-mini listwise reranker instead of Jev.

Full context. Every turn of the conversation, rendered as T0R renders a line, in the answer prompt.

engram v2. An LLM extracts facts from each message with mem0’s extraction prompt; Jev then answers typing questions and relation questions against up to ten candidate facts, and a belief policy closes superseded facts. The read path is T0R’s over facts instead of turns. We use the frozen v2 system (tag v2-frozen).

mem0 2.1.0. The default add() path: one LLM extraction call per message, with the session date as the observation date; reads are vector search.

Jev-Mem. Jev-Mem at commit 81574eb with its default profile and jev_model pinned to jev-1.13.0, driven through its own API. Each turn is a node; each write makes two Jev requests (memory type, relations), and each read routes, traverses and stops with between two and sixteen Jev requests. Its returned turns are rendered as “[date] speaker: text” and answered with our prompt; its own prompts, best-of-three selection and judge are not used.

A worked example. Figure 2 traces one held-out question through T0R and engram v2 at the H1 budgets. It was chosen by a fixed rule, not for effect: among the 8 temporal H1 questions that the judge and the human grader both scored correct for T0R and wrong for engram v2, the one with the shortest T0R context among those whose evidence turn Jev kept and whose replayed contexts match the recorded ones. engram v2 extracted the

evidence turn, but under the wrong speaker, and three other facts outranked it at $k=3$; T0R kept the verbatim turn with its date.

4 Study Design

Pre-registration. The plan was deposited before any run on the data below (10.5281/zenodo.22970745, commit b3c5dc5, tag v3-frozen). An amendment, with the outcome paragraphs used in §5.1, was deposited before any primary-test result was seen (10.5281/zenodo.22977848, commit efae0b6, tag v3-amended) (Sharma, 2026a,b).

Data. LoCoMo (Maharana et al., 2024) numbers its question categories. We name them 1 multi-hop, 2 temporal, 3 open-domain, 4 single-hop and 5 adversarial, which matches the dataset’s counts over all ten conversations (282, 321, 96, 841 and 446 questions). The primary data are conv-44, conv-47, conv-48, conv-49 and conv-50, never run by any system before this study: 3,122 turns, 778 scored questions (140 multi-hop, 165 temporal, 50 open-domain, 423 single-hop) and 209 adversarial. conv-30, conv-41, conv-42 and conv-43 (610 scored questions), held out in an earlier study, give an exploratory replication. Development used conv-26 only. LongMemEval_S cleaned (Wu et al., 2025) provides a registered sample of 70 questions (user turns only) and, by amendment, all 500 questions with user and assistant turns (470 scored, 30 abstention).

Stack. Answers and judgments use gpt-4o-mini at temperature 0 with mem0’s LoCoMo answer and judge prompts; the judge returns CORRECT or WRONG. Tokens are counted with o200k_base over the memory block the answer model sees.

Token matching. Every comparison between systems holds context fixed. The comparator runs at $k=3$, its natural setting, and T0R is matched to it: from a retrieval-only sweep of T0R over k from one to thirty, the k whose pooled mean tokens per question is closest to the comparator’s, ties going to the larger k . The sweep and the chosen k were saved before T0R answered at that k .

Tests. The primary test H1 asks whether T0R is non-inferior to engram v2: with d the per-question difference in correctness (T0R minus engram v2), non-inferiority holds if $\bar{d} - 1.645 \cdot \text{SE}$ exceeds -5 points. The margin is half the rerank’s measured effect on the development conversation. The seven secondary tests, under Holm correction at family-wise 0.05, are exact two-sided McNemar tests ex-

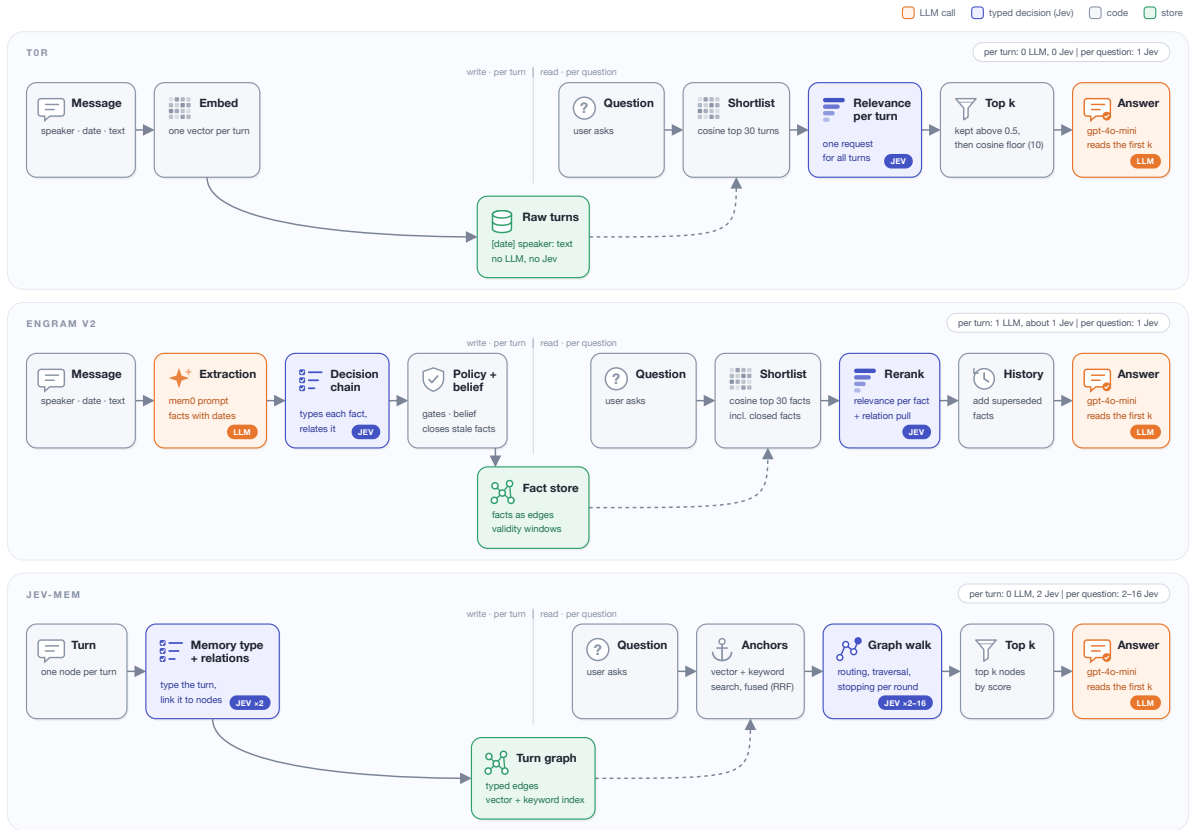


Figure 1: Write path (per turn) and read path (per question) of T0R, engram v2 and Jev-Mem. The outline colour says what does the work: an LLM call (orange), a Jev typed decision (blue), code (grey) or a store (green); dashed arrows are reads from the store. The chip on each panel gives the LLM calls and Jev requests per turn and the Jev requests per question, from each system’s code (Jev-Mem: its default profile). A design diagram; no measured data.

cept S4, a non-inferiority test with the same margin: S1 T0R against L0, S2 against Jev-Mem, S3 against mem0 and S4 against T0R-LLM on LoCoMo; S5 against mem0 and S6 against L0 on the LongMemEval sample; S7 against L0 on the full LongMemEval set. The plan’s power analysis put the probability of passing H1 at 0.89 if the development difference held.

Checks. H1 and S1 were re-answered by Llama 3.3 70B Instruct via OpenRouter from the same contexts and judged by the same judge; a result is called model-robust only if it holds under both answer models. The author graded every question on which the judge found exactly one of H1’s two answers correct, blind to system and judge label (§5.4, Appendix C). Shortlist recall measures where the LoCoMo evidence turns fall (§5.6, Appendix D).

Deviations. All are recorded in the plan with dates (Appendix B): the amendment; mem0’s extraction calls, which were served through OpenRouter (half of them by Azure) rather than the OpenAI API; runs repeated after rate limits; a token-

counting fix for one LongMemEval haystack; the audit sheet’s layout and grading standard; and the replacement of the title the registered outcome rule selected.

5 Results

5.1 Primary test (H1, registered)

The registered outcome paragraph, filled in:

Pass. At matched context (265 tokens; engram v2 251), raw turns with a single rerank call were non-inferior to LLM-extraction memory: difference -0.5 points, one-sided 95% lower bound -3.0 , above the registered -5 margin. Whatever accuracy extraction adds at this budget is under 3.0 points, at $3,061\times$ the write cost. The rerank closes 94% of the gap between similarity search and extraction. By category, T0R did not trail on multi-hop (74.3 vs 72.1, $n=140$) and trailed on open-domain (56.0 vs 60.0,

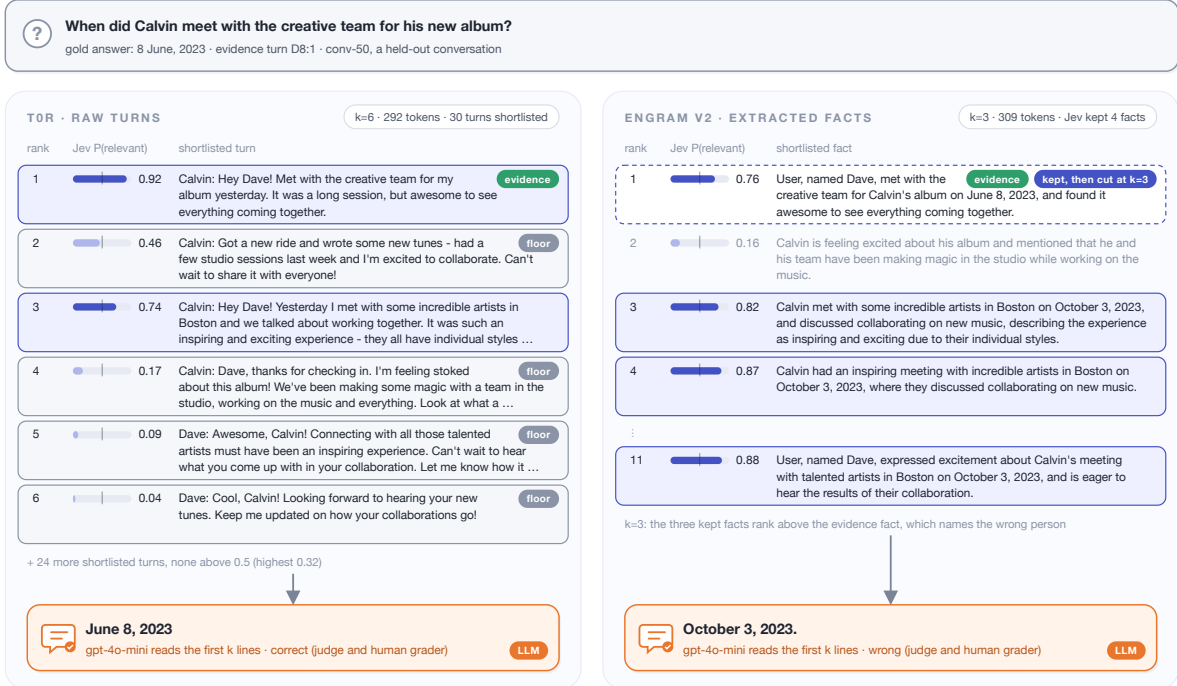


Figure 2: One held-out question traced through both read paths, replayed offline from the frozen stores and the call cache (no API call; both contexts match the recorded token counts: T0R 292, engram v2 309). Left: T0R’s 30-turn cosine shortlist, ranks shown in cosine order, with Jev’s $P(\text{relevant})$ (tick at the threshold); blue rows were kept by Jev, grey rows by the cosine floor, and the answer model reads the first $k=6$. Right: engram v2’s fact shortlist; Jev kept four facts, and the evidence fact (dashed) was cut at $k=3$. An illustration chosen by the rule in §3, not evidence.

Table 1: H1: T0R at $k=6$ (265 tokens per question) against engram v2 at $k=3$ (251 tokens), 778 scored questions of the five held-out conversations. The judge row is the registered test; the human rows replace the judge’s labels on the 141 graded discordant questions. Differences and bounds in points.

Grading	T0R	engram v2	Difference	One-sided 95% bound	Two-sided 95% CI	Non-inferior (margin -5)
Judge (registered)	77.0%	77.5%	-0.5	-3.0	[-3.5, +2.5]	yes
Human, strict	76.3%	78.0%	-1.7	-3.9	[-4.3, +0.9]	yes
Human, lenient	77.4%	79.9%	-2.6	-4.7	[-5.1, -0.03]	yes

$n=50$), contrary to what we registered for multi-hop and as we registered for open-domain.

The one-sided p-value is 0.0017, and the conversation bootstrap puts the fifth percentile of the difference at -2.3 points. 69 questions were answered correctly only by T0R and 73 only by engram v2. Human grading moves the difference to between -1.7 and -2.6 points and the bound to between -3.9 and -4.7; under lenient grading the two-sided interval lies just below zero, so by that grading engram v2 is more accurate, still inside the margin (Figure 3). We therefore state the result as: non-inferior within a 5-point margin under every grading we applied, with extraction adding at most

4.7 points at this budget. The category comparisons are descriptive; the categories are small and the differences are not tested.

G, the share of the gap between similarity search and extraction that the rerank closes, uses L0 at its own matched k (6, 68.6% accuracy): at the same token budget, one rerank call closes $G = 94\%$ of the accuracy gap between cosine retrieval of raw turns (L0, 68.6%) and LLM-extracted memory (engram v2, 77.5%). The write-cost ratio uses held-out measurements: engram v2 costs \$1.865 per 1,000 turns and T0R \$0.00061 (embeddings only), at list prices.

Table 2: Secondary tests. In each, the comparator runs at $k=3$ and T0R at its matched k . “Only T0R” and “only other” count questions answered correctly by one system. S4 is a non-inferiority test (one-sided p); the rest are exact two-sided McNemar tests. Holm adjustment over S1–S7. LoCoMo tests use 778 questions; LongMemEval ingestion: user turns for S5–S6, user and assistant turns for S7. Bold: rejected after Holm adjustment (family-wise 0.05).

Test	Comparison	T0R k	T0R	Other	Only T0R / only other	p	Holm p
S1	T0R vs L0 (LoCoMo)	3	77.2%	59.9%	152 / 17	$2.7e-28$	$1.9e-27$
S2	T0R vs Jev-Mem (LoCoMo)	4	77.0%	70.6%	98 / 48	$4.3e-5$	$1.3e-4$
S3	T0R vs mem0 (LoCoMo)	3	77.2%	68.5%	134 / 66	$1.7e-6$	$7.6e-6$
S4	T0R vs T0R-LLM (LoCoMo, non-inferiority)	3	77.2%	77.6%	28 / 31	$1.5e-6$	$7.6e-6$
S5	T0R vs mem0 (LongMemEval, 30 knowledge-update)	2	70.0%	70.0%	4 / 4	1.00	1.00
S6	T0R vs L0 (LongMemEval sample, 70)	3	68.6%	65.7%	7 / 5	0.77	1.00
S7	T0R vs L0 (LongMemEval, 470)	3	66.8%	57.7%	61 / 18	$1.3e-6$	$7.6e-6$

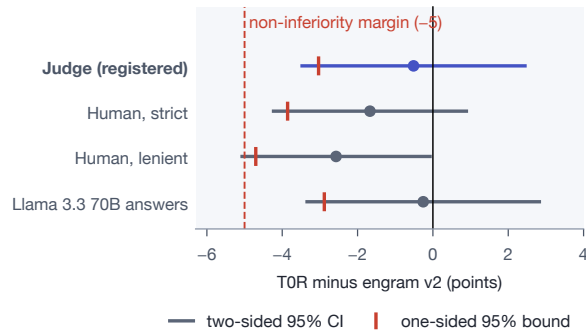


Figure 3: H1 (registered) as a forest plot: T0R at $k=6$ minus engram v2 at $k=3$, in points, on the 778 questions of the five held-out conversations. Bars are two-sided 95% intervals; the red tick is the one-sided 95% lower bound, tested against the -5 -point margin (dashed). The judge row is the registered test; the human rows replace the judge’s labels on the 141 graded discordant questions (§5.4); the Llama 3.3 70B row re-answers from the same contexts (the answer-model check of §4).

5.2 Secondary tests (S1–S7, registered)

S1, S2, S3, S4 and S7 are rejected after Holm correction; S5 and S6 are not. On LoCoMo, T0R was more accurate than similarity search (S1), Jev-Mem (S2) and mem0 (S3) at matched context, and non-inferior to the LLM reranker (S4: difference -0.4 points, lower bound -2.0). S3 carries a caveat: mem0’s extraction was served through OpenRouter, about half of it by Azure (Appendix G). On the LongMemEval sample, S5 detected no difference between T0R and mem0 on 30 knowledge-update questions, which is too few to establish equivalence, and S6 detected none between T0R and L0 on 70 questions. On the full set, S7 found T0R more accurate than L0 by $+9.1$ points. By the registered rule, LongMemEval holds: S7 favours T0R after Holm correction and S5 does not favour mem0. Figure 4

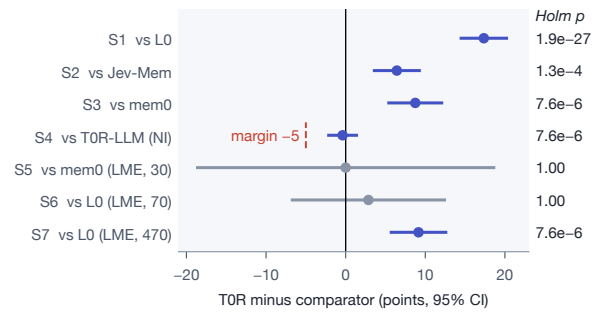


Figure 4: Secondary tests S1–S7 (registered): T0R minus the comparator, in points, with paired 95% intervals; the Holm-adjusted p is printed at the right, and blue rows are rejected after Holm correction. S4 is a non-inferiority test against the -5 -point margin (dashed). LoCoMo tests use 778 questions; S5 and S6 use the LongMemEval sample (30 and 70 questions), S7 the full set (470).

shows the paired differences with their intervals.

5.3 The budget dependence of reranking

The rerank’s value depends on how many candidates the budget keeps (Figure 5). On LoCoMo its gain over similarity search is $+17.4$ points at $k=3$ and $+1.5$ at $k=20$. On the full LongMemEval set it is $+9.1$ points at $k=3$ (S7) and $+1.1$ at $k=20$. With three of 30 candidates kept, ordering decides which evidence reaches the answer model; with twenty kept, cosine order already includes most of it. The $k=3$ gains are registered tests (S1, S7); the $k=20$ differences are descriptive.

T0R at $k=3$ is within 1.0 points of full context on LoCoMo while reading 139 tokens per question instead of 23,631. At generous budgets the ordering reverses (Table 3, Figure 6): engram v2 at $k=20$ was the most accurate system we measured (82.4% with 1,238 tokens), above Jev-Mem

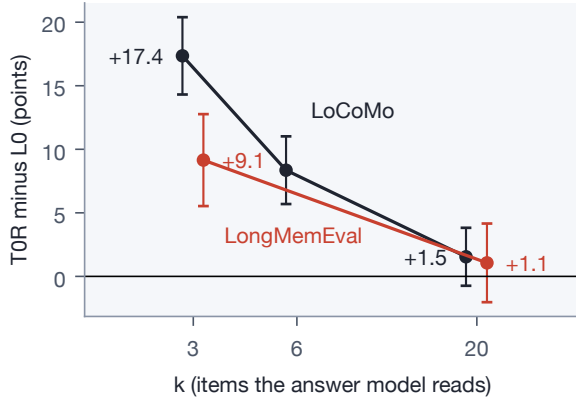


Figure 5: The rerank’s gain over similarity search (T0R minus L0, paired, in points, with 95% intervals) against k . LoCoMo: 778 questions of the five held-out conversations at $k=3$, $k=6$ and $k=20$; LongMemEval: 470 non-abstention questions, user and assistant turns, at $k=3$ and $k=20$. The $k=3$ points are registered tests (S1, S7); the others are descriptive.

at $k=40$ (80.3%, 1,987 tokens) and mem0 at $k=20$ (78.7%, 839 tokens); full context scored 78.3%, and T0R stays near 77.6% at any k (§5.6). This comparison is descriptive, not a registered test, and the systems are not token-matched.

5.4 Robustness: a second answer model and blind human grading

With Llama 3.3 70B Instruct answering from the same contexts, H1 still passes (T0R 75.3%, engram v2 75.6%, difference -0.3 , one-sided bound -2.9) and so does S1 (T0R 74.9%, L0 57.6%, $p = 5.8e-26$). Both are therefore model-robust by the registered rule. Of 3,112 rebuilt contexts, all but 4 matched their recorded token counts exactly; the four come from near-tie reorderings. OpenRouter served Llama through 9 providers whose numeric precision may differ.

The author graded, blind, both answers to each of H1’s 142 judge-discordant questions (284 rows; one question was left ungraded). Agreement with the judge was 81% under the strict mapping and 79% under the lenient one (Appendix C). Many judge-discordant pairs were not discordant to the human grader: under the strict mapping both answers were correct for 17 questions and both wrong for 18. The judge credited T0R’s short answers more readily and engram v2’s list-style answers less (agreement on engram v2’s answers 77% under the lenient mapping, against 82% on T0R’s), which is why human grading widens the gap.

5.5 Long histories (LongMemEval)

LongMemEval compares T0R with mem0 and L0 only; engram v2 was not run on it, so these results cannot support any claim that T0R matches LLM-extracted memory on long histories. What they support is narrower: on histories of about 111,770 rendered tokens, the rerank still beats similarity search (S7), and no difference from mem0 was detected on knowledge-update questions (S5).

Full context scored 63.0% against T0R’s 66.8% at $k=3$ (72 questions correct only for T0R and 54 only for full context, $p = 0.13$, descriptive), reading $209\times$ the tokens at $29\times$ the cost per question (\$0.0169 against \$0.00058 for reading and answering, judge excluded). It was weakest on temporal and multi-session questions. On the registered sample (user turns only), T0R scored 68.6% at $k=3$ and L0 65.7%; mem0 scored 70.0% on the knowledge-update questions at $k=3$.

5.6 Where T0R’s accuracy stops

T0R levels off near 77.6%: at $k=20$ it reads only 496 tokens, because its rerank keeps only short-listed turns scored above 0.5. Shortlist recall (exploratory as registered) locates the loss on all nine held-out conversations (1,388 questions). All evidence turns were in the 30-turn shortlist for 76.9% of questions and at least one for 88.5%; among questions with evidence in the shortlist, the rerank kept none of it for 10.4%. About 11.5% of questions are lost to the shortlist and another 9.2% to the rerank.

Figure 7 shows the same decomposition by category on the five held-out conversations.

The categories differ. Open-domain evidence reaches the shortlist least often (63.0%) and is dropped most often (31.4%), consistent with T0R trailing engram v2 on open-domain questions. Temporal evidence usually reaches the shortlist but is dropped by the rerank for 22.7% of questions: a turn that only establishes when something happened does not look relevant to the question on its own. Multi-hop questions usually get some evidence into the shortlist (88.4%) but rarely all of it (42.8%).

A wider read path (post-hoc exploratory). After the registered results were in, we tested one variant once, outside the Holm family and in its own ledger: T0R-wide takes a 150-turn cosine shortlist, asks Jev about every shortlisted turn, and keeps the top k by Jev’s score with no cut-off, with

Table 3: LoCoMo, five held-out conversations, 778 scored questions: accuracy (%), tokens per question, write cost per 1,000 turns and read cost per query at list prices (as in Table 5; the read cost excludes the answer call; “–”: none), and accuracy by category. Rows are grouped by budget. Bold: best in column within the budget group (highest accuracy, lowest cost).

System, setting	Accuracy	Tokens	Write \$/1k turns	Read \$/query	Multi-hop	Temporal	Open-domain	Single-hop
<i>Tight budget ($k=3$ or $k=6$)</i>								
L0, $k=3$	59.9%	130	\$0.0006	\$0.00000	54.3	55.2	42.0	65.7
L0, $k=6$	68.6%	254	\$0.0006	\$0.00000	61.4	60.6	54.0	75.9
T0R, $k=3$	77.2%	139	\$0.0006	\$0.00020	75.7	69.1	58.0	83.2
T0R, $k=6$	77.0%	265	\$0.0006	\$0.00020	74.3	72.1	56.0	82.3
T0R-LLM, $k=3$	77.6%	143	\$0.0006	\$0.00023	73.6	72.7	58.0	83.2
engram v2, $k=3$	77.5%	251	\$1.865	\$0.00018	72.1	77.6	60.0	81.3
mem0, $k=3$	68.5%	129	\$1.321	\$0.00000	56.4	67.3	56.0	74.5
Jev-Mem, $k=3$	70.6%	162	\$0.212	\$0.00120	57.9	64.2	50.0	79.7
<i>Generous budget ($k=20$ or $k=40$, and full context)</i>								
L0, $k=20$	76.1%	826	\$0.0006	\$0.00000	68.6	68.5	58.0	83.7
T0R, $k=20$	77.6%	496	\$0.0006	\$0.00020	75.0	72.7	58.0	82.7
T0R-LLM, $k=20$	78.0%	456	\$0.0006	\$0.00023	72.9	71.5	60.0	84.4
engram v2, $k=20$	82.4%	1,238	\$1.865	\$0.00018	77.9	80.0	64.0	87.0
mem0, $k=20$	78.7%	839	\$1.321	\$0.00000	74.3	76.4	54.0	83.9
Jev-Mem, $k=40$	80.3%	1,987	\$0.212	\$0.00174	76.4	73.9	54.0	87.2
Full context	78.3%	23,631	–	–	78.6	57.0	64.0	88.2

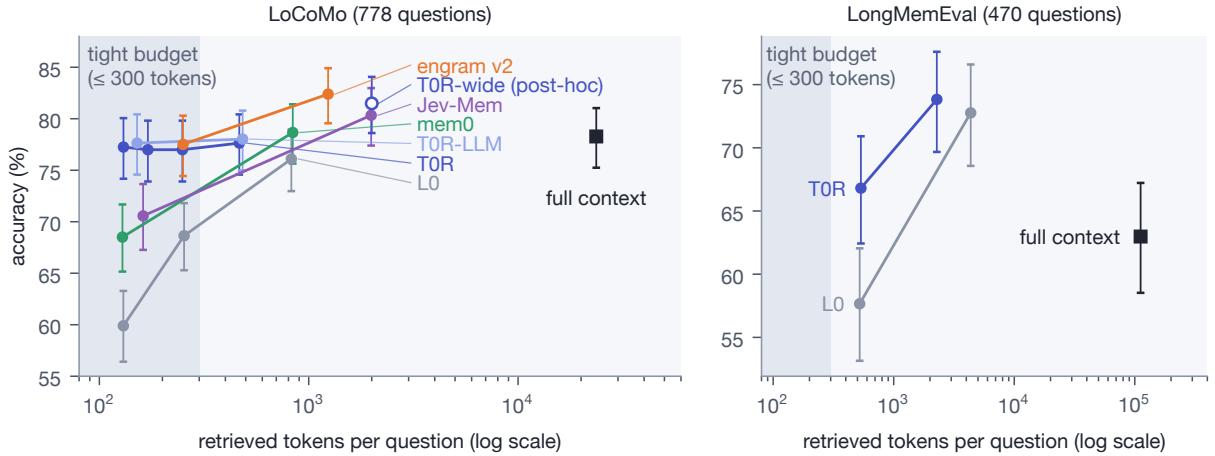


Figure 6: Accuracy against retrieved tokens per question (log scale), with Wilson 95% intervals. Left: LoCoMo, 778 questions of the five held-out conversations, each system at each k it was run. Right: LongMemEval, 470 non-abstention questions, user and assistant turns. Shaded: the tight budget (at most 300 tokens). T0R-wide (hollow) is post-hoc; the other points are registered runs, compared descriptively except in the tests of §5.1–§5.3.

Table 4: LongMemEval, all 500 questions, user and assistant turns ingested: accuracy (%) on the 470 non-abstention questions and by question type (n in parentheses), and the share of the 30 abstention questions answered by abstaining. KU: knowledge update; MS: multi-session; SS-A, SS-P, SS-U: single-session assistant, preference and user; TR: temporal reasoning; Abs: abstention. Bold: best in column within the budget group.

System	Accuracy	Tokens	KU (72)	MS (121)	SS-A (56)	SS-P (30)	SS-U (64)	TR (127)	Abs
<i>Tight budget ($k=3$)</i>									
L0, $k=3$	57.7%	522	58.3	31.4	85.7	53.3	95.3	52.0	43.3%
T0R, $k=3$	66.8%	534	77.8	47.9	98.2	46.7	98.4	53.5	43.3%
<i>Generous budget ($k=20$, and full context)</i>									
L0, $k=20$	72.8%	4,352	84.7	58.7	98.2	40.0	96.9	63.8	70.0%
T0R, $k=20$	73.8%	2,278	84.7	67.8	96.4	46.7	96.9	58.3	63.3%
Full context	63.0%	111,770	81.9	45.5	91.1	56.7	92.2	43.3	70.0%

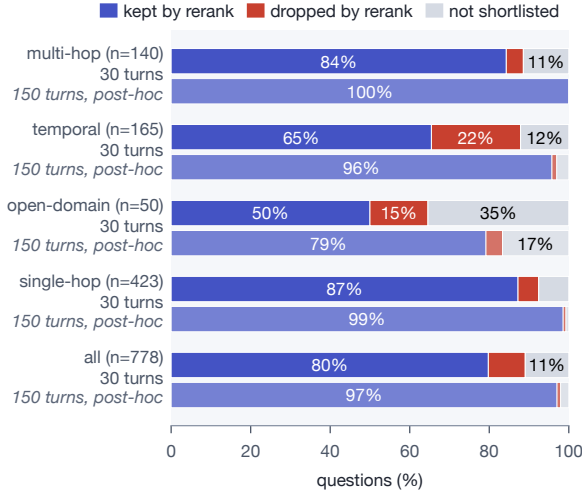


Figure 7: Where T0R’s accuracy stops, by category, on the 778 questions of the five held-out conversations: the rerank kept at least one evidence turn (blue), evidence was in the shortlist but the rerank kept none of it (red), or no evidence turn was in the shortlist (grey). Upper bar of each pair: T0R’s 30-turn shortlist (shortlist recall, exploratory as registered). Lower, lighter bar: T0R-wide’s 150-turn shortlist (post-hoc). Percentages are printed where the segment is wide enough.

$k=47$ matched to Jev-Mem at $k=40$ (2,000 tokens against 1,987). It scored 81.5%, against 80.3% for Jev-Mem at $k=40$ and 82.4% for engram v2 at $k=20$ (1,238 tokens); all-evidence recall rose to 91.9% and the rerank’s losses fell to 0.9% (Figure 7, lower bars). This suggests the ceiling comes from T0R’s read path rather than from storing raw turns, a hypothesis for new data, not a finding of this study.

5.7 Cost and latency

Jev reads about $3.0\times$ faster than the LLM reranker and $4.9\times$ faster than Jev-Mem at $k=3$; engram v2 reads as fast as T0R because its read path is the same one request. The write cost is where the systems differ: extraction makes engram v2 and mem0 thousands of times more expensive to write than T0R, and Jev-Mem’s two Jev requests per turn cost \$0.211 per 1,000 turns. Jev-Mem at $k=40$ averaged 3.0 Jev calls per query (up to 11) and \$0.00174 per query. Mem0’s read cost is a query embedding only.

Figure 8 amortises write cost at the benchmark’s own ratio (4.0 turns written per scored question). At that ratio T0R’s total cost per question is \$0.00030 and engram v2’s \$0.00778; full context costs \$0.00362. In a read-heavy use with one turn written per question, the write cost weighs less:

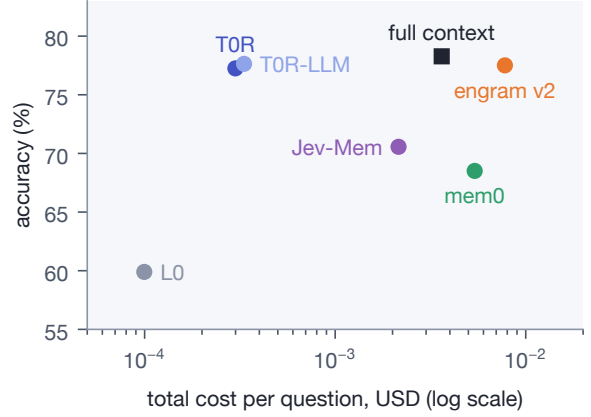


Figure 8: Accuracy against total cost per question (log scale) at $k=3$, on the 778 questions of the five held-out conversations (exploratory as registered): write cost amortised at the benchmark’s 4.0 turns written per question, plus read cost and answer cost (judge excluded), at list prices; full context has no write or read cost. In a read-heavy use with one turn written per question, engram v2’s total falls to \$0.00216, mem0’s to \$0.00142 and Jev-Mem’s to \$0.00151; the other systems’ totals do not change at this precision.

engram v2’s total falls to \$0.00216.

5.8 Abstention

Reranking lowered correct abstention at $k=3$, with Jev (54.1%) and with an LLM reranker (48.8%) against similarity search (63.6%): relevant-looking context makes the answer model less willing to say that something was not mentioned. The exploratory conversations show the same (54.2% against 65.3%), and so does LongMemEval at $k=20$ (63.3% for T0R against 70.0% for L0). Fidelity Before Structure reports that verbatim chunks abstain worse than extracted artifacts; we find that reranking specifically adds to that.

6 Discussion

A budget reading of two prior findings (interpretation). SmartSearch finds ranking to be the bottleneck; Fidelity finds reranking marginal. Our within-study results suggest the difference is the budget: reranking matters in proportion to how hard truncation cuts the candidate set. In SmartSearch a question has about 431 grep candidates on average, of which about 62 passages fit its 2,000-word budget, and without ranking only 22.5% of gold evidence survives truncation; our $k=3$ keeps three of 30. Both show large ranking gains. Fidelity reranks a top-30 pool to 15 with bge-reranker-v2-m3 under a 5,000-token cap (gains of 2.9 points

Table 5: Cost and latency (exploratory as registered), all at $k=3$ (the tight budget). Write cost per 1,000 turns at list prices, split by LLM, Jev and embeddings; read cost per query; read latency measured live on a fixed sample of 40 questions at $k=3$, one query at a time, including the query-embedding call. Jev-Mem’s latency comes from its reads of the same questions, measured live when they ran (42 reads: two question texts repeat). A write-latency range spans the conversations and is compared by its lower end. Bold: best in column within the budget group (lowest cost and latency).

System	Write \$/1k turns	LLM	Jev	Embeddings	Write p50 (s)	Read \$/query	Jev calls/query	Read p50 (ms)	Read p90 (ms)
T0R	\$0.0006	–	–	\$0.0006	0.2	\$0.00020	one	273	359
L0	\$0.0006	–	–	\$0.0006	0.2	\$0.00000	none	219	263
T0R-LLM	\$0.0006	–	–	\$0.0006	0.2	\$0.00023	one (no-op)	816	1,107
engram v2	\$1.865	\$1.322	\$0.542	\$0.0015	2.0–2.1	\$0.00018	one	276	319
mem0	\$1.321	\$1.319	–	\$0.0017	2.1–2.3	\$0.00000	none	486	718
Jev-Mem	\$0.212	–	\$0.211	\$0.0011	0.5	\$0.00120	5.3	1,329	1,622

Table 6: Share of LoCoMo adversarial questions (209, five held-out conversations) answered by abstaining (exploratory as registered). Each column is a budget group. Bold: best in column within the budget group (highest share).

System	$k=3$	$k=20$
L0	63.6%	47.8%
engram v2	59.8%	52.2%
mem0	59.8%	53.6%
T0R	54.1%	49.3%
T0R-LLM	48.8%	53.1%

on LoCoMo and 0.6 on LongMemEval-S), and our $k=20$ keeps twenty of 30; both show small ones. This is an interpretation across pipelines that differ in retrievers, rerankers, answer models and judges, supported inside our study by the $k=3$ to $k=20$ comparison on two benchmarks, not a tested claim across papers.

When extraction is worth it. At the tight budget of H1, extraction adds little and costs thousands of times more to write. At generous budgets it is more accurate and more compact: engram v2 at $k=20$ was the most accurate system we measured, with fewer tokens than Jev-Mem at $k=40$. Open-domain and temporal questions are where T0R trailed engram v2, and where its shortlist and rerank lose the most evidence.

What a typed decision model contributes. In this study, speed and cost at equal selection quality, not higher accuracy: Jev selected as accurately as the gpt-4o-mini reranker (S4) at about a third of the latency, in one request per question, and one Jev request beat Jev-Mem’s multi-request graph walk at matched context (S2). A typed question returns a probability over fixed options in one short call, which is what a reranker needs.

Judge leniency and answer style. mem0’s LoCoMo judge is lenient, and in our audit it credited short answers more readily than list-style ones, so its agreement with human grading differed by system. Fidelity Before Structure’s human study found no such dependence on answer length, but its LoCoMo judge is instructed to be strict (“Binary - strict”: “partial answers or answers with significant missing information should be marked INCORRECT”, their Appendix J.4), while mem0’s asks the judge to “be generous with your grading - as long as it touches on the same topic as the gold answer”. A strict instruction leaves less room for style to matter, which may explain why their audit found no short-answer bias and ours did. Memory benchmarks that compare systems with different answer styles should report judge–human agreement by system.

Not state of the art. SmartSearch reports 91.9% on LoCoMo under its own protocol (gpt-4o-mini answering and judging, binary judgments, all ten conversations and 1,540 questions in categories 1–4, 3,141 tokens per question); our numbers come from a different protocol and five held-out conversations and are not comparable to it. Our best result, the post-hoc T0R-wide, is below that figure.

Limitations

- **Benchmarks.** One benchmark family per setting: LoCoMo, whose dialogues are LLM-generated, and LongMemEval. Five primary conversations give 778 questions; categories are small.
- **LongMemEval scope.** engram v2 was not run on LongMemEval, so no claim about extraction on long histories follows from this study.
- **Human grading.** The grader is the system’s

author; the mapping of partial grades was not pre-specified (two are reported); one question was ungraded; and only judge-discordant questions were re-graded, so judge errors on questions where the judge agreed across systems remain.

- **A closed decision model.** Jev is a closed, versioned model; results hold for jev-1.13.0.
- **mem0 serving.** mem0’s extraction calls were served through OpenRouter, about half by Azure, not the OpenAI API as registered; only S3 involves mem0.
- **Post-hoc variant.** T0R-wide was designed after the registered results and tested once on the same questions.
- **Absolute accuracy.** Below SmartSearch’s reported figures at generous budgets, under a different protocol.
- **Development data.** Every design choice was made on one conversation, conv-26.

7 Conclusion

The context budget explains much of the published disagreement about conversational memory. Selecting raw turns matters when the budget keeps a few of many candidates: reranking added 17.4 points on LoCoMo and 9.1 on LongMemEval when three of 30 were kept, and 1.5 and 1.1 at $k=20$, where extraction systems were more accurate. A pre-registered non-inferiority test on held-out conversations, with a second answer model and blind human grading, bounds what extraction adds at a tight budget to at most 4.7 points, at $3,061\times$ the write cost. A typed decision model is an effective selector: Jev was non-inferior to an LLM reranker (bound -2.0) at about a third of the latency and more accurate than a multi-call Jev graph traversal at matched context. The diagnostics locate where selection stops, in shortlist misses (11.5%) and rerank drops (9.2%), most for temporal evidence, and show that an LLM judge’s leniency interacts with answer length, so memory benchmarks that compare systems with different answer styles should report judge–human agreement by system.

AI Assistance

The code, run orchestration and drafting of this paper were done with Claude Code (Anthropic) under the author’s direction. The author made every

methodological decision, approved each stage of the registered plan and did the human audit.

Artifacts

Code, plans, per-question answers and judge labels, and the human-audit grades with their key are at github.com/ris3abh/Engram: tags v3-frozen, v3-amended and the paper tag; results in `bench/results/v3/` (per-question files, reports, ledgers, human_audit/) and `bench/results/v3_posthoc/`. The plan and its amendment are deposited at 10.5281/zenodo.22970745 and 10.5281/zenodo.22977848; the earlier engram preprint is 10.5281/zenodo.22941757 (Sharma, 2026c).

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A Per-conversation results

Table 7: Accuracy (%) per conversation, LoCoMo scored categories.

System, setting	conv-44 (123)	conv-47 (150)	conv-48 (191)	conv-49 (156)	conv-50 (158)
T0R, $k=3$	78.0	76.7	79.1	73.7	78.5
T0R, $k=6$	79.7	76.7	78.5	73.7	76.6
T0R, $k=20$	79.7	74.7	81.7	75.0	76.6
L0, $k=3$	65.9	57.3	62.8	58.3	55.7
L0, $k=20$	78.0	74.0	79.1	74.4	74.7
T0R-LLM, $k=3$	76.4	72.7	80.6	77.6	79.7
engram v2, $k=3$	77.2	80.7	78.5	76.3	74.7
engram v2, $k=20$	84.6	84.7	82.2	80.8	80.4
mem0, $k=3$	68.3	65.3	69.6	72.4	66.5
mem0, $k=20$	79.7	78.0	81.2	78.8	75.3
Jev-Mem, $k=3$	69.1	66.7	74.3	69.2	72.2
Jev-Mem, $k=40$	81.3	78.7	80.6	82.7	78.5
Full context	82.1	72.7	82.7	76.3	77.2

The exploratory replication on conv-30, conv-41, conv-42 and conv-43 (610 scored questions): L0 60.7% at $k=3$ and 76.2% at $k=20$; T0R 77.2% at $k=3$ and 76.6% at $k=20$. T0R’s matched k against L0 was 3, with 116 questions correct only for T0R and 15 only for L0 ($p = 1.6e-20$, exploratory).

B Registered plan and deviations

The plan (its guarded sections in docs/V3_PLAN.md, checked by a test that fails on any undated change, fixed the systems, data, token-matching rule, tests, predictions, human check, run order and budget before any run. Every change is a dated entry in its Deviations section:

- **2026-09-26, amendment**, deposited before any primary-test result was seen: shortlist recall; Long-MemEval on all 500 questions with user and assistant turns and the new test S7; the second answer model; the outcome paragraphs and the rule for “LongMemEval holds”; budget caps.
- **2026-09-26, outcome paragraphs revised** before upload, before any primary-test result was seen.
- **2026-09-26, Batch B execution**: mem0’s extraction was routed to OpenRouter by a library default ([Appendix G](#)); runs hit OpenAI’s rate limit and were repeated with more retries and a Jev throttle, completed calls replaying from a call cache; T0R-LLM also asks Jev one query-relation question per query, a no-op on turns.
- **2026-09-26, mem0 via OpenRouter resolved**: model and serving provider established, billed amount corrected in the ledger, guards added before any later run.
- **2026-09-27, human check layout** (record only): each answer on its own row, all 284 rows shuffled together, rather than randomising the order within each question.
- **2026-09-27, human check grading standard** (record only): the sheet asked for CORRECT or WRONG; partial grades appeared, so two mappings are reported.
- **2026-09-27, paper title** (presentation change): the title the outcome rule selected, “Selection, Not Extraction: One Rerank Call Matches LLM-Extracted Memory at a Fraction of the Write Cost”, was replaced because it presents a published idea as new and its “matches” overstates a non-inferiority result.
- **2026-09-26, token counting** (record only): one LongMemEval haystack contains the literal text `<|endof text|>`, which the token counter refused; counting now treats it as ordinary text, the affected question was re-run, and every other count is unchanged.

An implementation note not in the Deviations section: two retrieval-only sweeps were stopped by mistake and re-run from the cache.

C Human audit

Protocol. The sheet held every question on which the judge found exactly one of H1’s two answers correct (142 questions), each answer as its own row (284 rows), shuffled with seed 0, with the question and gold answer shown and no system name or judge label. The sheet asked for CORRECT or WRONG; the plan’s notes had listed CORRECT, WRONG or UNCLEAR. The author’s grades included partial and hedged labels, and one question was left ungraded. Two mappings are reported: strict (only grades starting with CORRECT count as correct) and lenient (partial and hedged-correct grades also count); any grade containing WRONG counts as wrong under both. H1 is decided by the judge.

Mapping	Agreement with judge	On T0R’s answers	On engram v2’s answers	Only T0R right	Only engram v2 right	Both right	Both wrong
Strict	81%	81%	82%	47	59	17	18
Lenient	79%	82%	77%	41	60	31	9

The grades, the key and the analysis are in `bench/results/v3/human_audit/` and `bench/v3_human_audit.py`.

D Shortlist recall

For every scored question of the nine held-out conversations, the 30-turn cosine shortlist (shared by L0 and T0R) and the turns T0R’s rerank keeps were rebuilt from the frozen stores through the call cache. Each turn’s id is its LoCoMo dialogue id, so the question’s evidence ids can be located. Reported: the share of questions with all, and with at least one, evidence turn in the shortlist, and, among questions with an evidence turn in the shortlist, the share where the rerank keeps none. Recall is scored on raw turns only (Panthi and Abdelfattah, 2026). The five fresh conversations (76.9% all-evidence recall, 10.4% dropped by the rerank) and the four exploratory ones (76.9%, 10.3%) agree.

Category	Questions	All evidence in shortlist	At least one	Rerank keeps none
Multi-hop	250	42.8%	88.4%	5.9%
Temporal	284	83.5%	88.4%	22.7%
Open-domain	83	42.0%	63.0%	31.4%
Single-hop	771	89.2%	91.2%	5.8%
All	1,388	76.9%	88.5%	10.4%

E T0R-wide (post-hoc exploratory)

Designed after the registered results were seen, tested once on the 778 questions of the five held-out conversations, outside the Holm family, in its own ledger (\$0.31 OpenAI and \$0.58 Jev). Design: T0R’s store; a 150-turn cosine shortlist; Jev’s relevance question on every shortlisted turn, thirty per request; the top k by Jev’s probability, with no cut-off, floor or expansion; $k=47$ matched to Jev-Mem at $k=40$ by the registered rule, saved before answering. Results: accuracy 81.5% at 2,000 tokens (multi-hop 78.6, temporal 76.4, open-domain 64.0, single-hop 86.5); read cost \$0.00089 per query; live read latency 720 ms at p50 and 776 ms at p90. With the same recall method on the 150-turn shortlist, all evidence was in the shortlist for 91.9% of questions and at least one turn for 97.9%, and the top k kept none of the shortlisted evidence for 0.9%.

F Prompts and the Jev question

The answer prompt is mem0’s LoCoMo answer prompt adapted to one memory list, and the judge is mem0’s LoCoMo accuracy prompt (`bench/locomo_subset.py`, `ANSWER_PROMPT` and `ACCURACY_PROMPT`). LongMemEval questions carry their question date in the question slot. T0R’s rerank asks Jev one yes/no question per shortlisted turn (`src/engram/decide/questions.py`, `RELEVANT_TO_QUERY`):

- instructions: "Does memory help answer query?"
- true: "It states or directly implies part of the answer."
- false: "It is off-topic or only shares a keyword."

G The mem0 serving incident

mem0 2.1.0 sends its OpenAI LLM calls to OpenRouter whenever an OpenRouter key is present in the environment, without warning (`mem0/llms/openai.py`). A key added for the second answer model therefore routed all 3,122 of mem0's extraction calls through OpenRouter, which served 1,599 of them by OpenAI and 1,523 by Azure, all as gpt-4o-mini, the registered model. OpenRouter billed \$2.42; at OpenAI list price the same calls cost \$4.12. S3's difference is far larger than a serving difference could explain, and S3 is reported with this caveat. Before any later run, mem0 was pinned to the OpenAI endpoint and the spend guard was extended to OpenRouter. Anyone benchmarking mem0 2.1.0 with an OpenRouter key in their environment will get routed calls without warning.

H Cost accounting

Every system's gpt-4o-mini, embedding and Jev calls are priced at list price (gpt-4o-mini \$0.15 and \$0.60 per million input and output tokens; Jev \$0.042 per million input tokens), so no system looks cheaper because of a discount the others did not get. Billed amounts are reported separately: the registered ledger records \$23.16 of OpenAI, \$5.55 of Jev and \$0.31 of OpenRouter for the second answer model (at \$0.10 and \$0.32 per million input and output tokens), plus mem0's \$2.42 through OpenRouter. Write-cost ratios use the held-out measurements; read costs are per query; totals per question state their reads-per-write assumption (Figure 8).

I Reproduction

Each table and figure is rebuilt from the committed result files, with no API calls:

- numbers: `uv run --extra bench python paper_v3/make_numbers.py`
- figures: `uv run --with matplotlib --with pillow python paper_v3/figures.py` (Figures 1 and 2 are SVGs from `paper_v3/diagram.py`, printed to PDF with headless Chrome)
- the worked example of Figure 2: `bench/v3_worked_example.py`, which replays both read paths from the frozen stores through the call cache opened read-only (a cache miss stops it, so it cannot call an API)
- paper: `make -C paper_v3 paper` (renders `main.md` and `main.tex`, builds the PDF, runs `paper_v3/check.py`)
- reports behind the tables: `bench/v3_report.py` (Batches A–C), `bench/v3_human_audit.py`, `bench/v3_shortlist_recall.py`, `bench/v3_latency.py`, `bench/v3_second_model.py`, `bench/v3_posthoc.py`

The runs themselves are `bench/run.py` with `--study v3`, `bench/jevmem_run.py` and `bench/v3_batch_c.py`, from tag v3-frozen onward, as recorded in the plan.